

Problem 15.28

A loan of \$10,000 is being repaid by 10 semiannual payments, with the first payment made one-half year after the loan. The first 5 payments are K each, and the final 5 are $K + 200$ each. What is K if $i^{(2)} = 0.06$?

Solution

Since $i^{(2)} = 0.06$, the effective semiannual interest rate is

$$j = \frac{0.06}{2} = 0.03.$$

The present value of the first five payments of K is

$$Ka_{\overline{5}|j}.$$

The final five payments of $K + 200$ occur one half-year period after the first five payments, so their present value at the loan date is

$$(K + 200)a_{\overline{5}|j}(1.03)^{-5}.$$

Therefore,

$$10000 = Ka_{\overline{5}|j} + (K + 200)a_{\overline{5}|j}(1.03)^{-5}.$$

Expanding,

$$10000 = K(a_{\overline{5}|j} + a_{\overline{5}|j}(1.03)^{-5}) + 200a_{\overline{5}|j}(1.03)^{-5}.$$

Solving for K ,

$$K = \frac{10000 - 200a_{\overline{5}|j}(1.03)^{-5}}{a_{\overline{5}|j} + a_{\overline{5}|j}(1.03)^{-5}}.$$

Using

$$a_{\overline{n}|j} = \frac{1 - (1 + j)^{-n}}{j},$$

we get

$$a_{\overline{5}|j} = \frac{1 - (1.03)^{-5}}{0.03} \approx 4.5797.$$

Thus,

$$\begin{aligned} K &= \frac{10000 - 200(4.5797)(1.03)^{-5}}{4.5797 + 4.5797(1.03)^{-5}} \\ &\approx 1079.6813. \end{aligned}$$

Hence,

$$\boxed{K \approx \$1,079.68}.$$